The Resonant LLC Converter

Overview, Voltage Gain, and Operating Modes

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1. Application Background and Topology Selection

The resonant LLC converter offers excellent characteristics such as **high efficiency, low EMI, and high power density**. Compared with PWM converters such as the buck and the boost, however, it is more difficult to design, the core challenge being the **optimal design of the resonant tank**. These notes are based on Infineon's *Resonant LLC Converter: Operation and Design*, AN 2012-09.

1.1 Why LLC Is Needed: Starting from the Hard Switching of PWM

Conventional PWM converters (buck, boost, forward, flyback, full bridge, etc.) generally suffer from **hard switching**: at the instant of turn-on/turn-off, the switch voltage and current are simultaneously nonzero, producing switching loss. This keeps the switching frequency f_{sw} from being pushed very high; and when the frequency cannot rise, the magnetic components and capacitors are hard to shrink, so the **power density** is limited.

The core idea of LLC is to use **resonance** to make the switch turn on **when the voltage is zero (ZVS, Zero Voltage Switching)**, thereby almost eliminating turn-on loss. The converter can then operate at high frequency → a small, high-density implementation.

PWM controls the output by **adjusting the duty cycle**; LLC controls the output by **adjusting the switching frequency** f_{sw}, and achieves soft switching through resonance.

2. Resonance

Since the behaviour of the LLC is entirely built on resonance, we first review the concept of resonance from circuit theory.

An inductor stores magnetic-field energy $\frac{1}{2}Li^2$ and a capacitor stores electric-field energy $\frac{1}{2}Cv^2$. Connecting the two, energy shuttles back and forth between the magnetic and electric fields, forming an oscillation. This natural oscillation frequency is the **resonant frequency**:

$$\omega_0 = \frac{1}{\sqrt{LC}}, \qquad f_0 = \frac{1}{2\pi\sqrt{LC}}. \tag{1}$$

Driving a series R–L–C with a sinusoidal source, the total impedance is

$$Z(\omega) = R + j\omega L + \frac{1}{j\omega C} = R + j\underbrace{\left(\omega L - \frac{1}{\omega C}\right)}_{\text{reactance } X}. \tag{2}$$

At the resonant point f_0, the reactance vanishes and $Z = R$ takes its **minimum value**. If R is very small, the impedance of this series branch approaches zero, **behaving like a plain wire (a short circuit)**, and the current reaches its maximum.

Table 1: Behaviour of the series RLC at different frequencies

Frequency	**Dominant term**	**Reactance X**	**Behaviour / current phase**
$f < f_0$	capacitive $\frac{1}{\omega C}$	$X < 0$	capacitive / current **leads** voltage
$f = f_0$	the two cancel	$X = 0$	purely resistive / current **in phase** with voltage
$f > f_0$	inductive ωL	$X > 0$	inductive / current **lags** voltage

2.1 The Quality Factor Q and Bandwidth

The quality factor Q measures the "sharpness" of the resonance (it also equals the ratio of the stored energy at resonance to the energy dissipated per cycle):

$$Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}, \qquad \Delta f = \frac{f_0}{Q}. \tag{3}$$

The smaller R is $\rightarrow$ the larger Q $\rightarrow$ the taller and sharper the curve and the narrower the bandwidth. **The Q in LLC has exactly this same form, only with R replaced by the reflected load resistance R_{ac}**, so "the heavier the load $\rightarrow$ the smaller R_{ac} $\rightarrow$ the larger Q".

The behaviour of the LLC is more complex than that of a simple series resonance, precisely because it **contains two resonances at once**: the series resonance of L_r with C_r, and the lower resonance of (L_r+L_m) with C_r when the secondary is disconnected. A series RLC has minimum impedance at resonance; a parallel RLC has maximum impedance at resonance (tending to an open circuit).

3. LLC Topology and Equivalent Circuit

As in Fig. 1, a full-bridge LLC converter (with full-bridge rectification) divides, from left to right, into three parts: the switching bridge chops the DC into a square wave, the square wave excites the LLC resonant tank, the tank outputs a resonant sinusoidal current, which is scaled and rectified through the transformer and rectifier, and finally filtered by the output capacitor to produce a DC voltage.

L_r is the resonant inductor (often provided by the transformer **leakage inductance**), L_m is the transformer **magnetizing inductance** (note: in PWM one usually wants the magnetizing inductance as large as possible and treats it as a parasitic, whereas in LLC it is a **design element that actively participates in the resonance**), and C_r is the resonant capacitor (which also provides DC blocking).

The resonant tank is a **frequency-selective network**, letting essentially only the energy at the **fundamental frequency** pass. Therefore, even though the switching bridge feeds in a square wave, the tank current I_{Lr} is approximately sinusoidal. On this basis, only the fundamental component need be analyzed — the **First Harmonic Approximation (FHA)**. This step reduces the complex switching circuit to a familiar linear RLC AC circuit (Fig. 2): the square-wave source is equivalent to a sinusoidal source V_{in_ac}, and the "transformer + rectifier + load" on the output side is equivalent as a whole to an AC resistance R_{ac}.

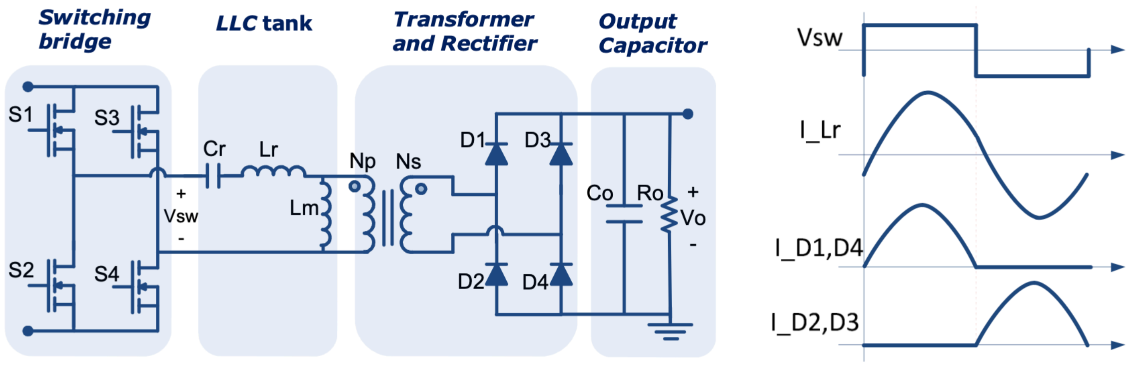

Figure 1: The full-bridge LLC resonant converter

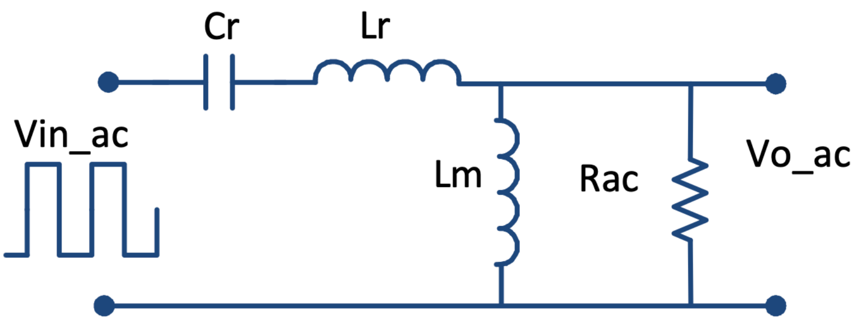

Figure 2: The equivalent resonant circuit under the first harmonic approximation

4. Converter Voltage Gain

The total gain of the converter is the product of three stages:

$$\text{total gain} = \underbrace{(\text{bridge gain})}_{\text{full=1, half=0.5}} \times \underbrace{K}_{\text{tank gain}} \times \underbrace{\frac{N_s}{N_p}}_{\text{turns ratio}} . \tag{4}$$

Here the bridge gain is 1 for a full bridge and 0.5 for a half bridge; the turns ratio N_s/N_p is a transformer parameter; and **the only quantity that varies with the switching frequency is the tank gain K.**

4.1 Transfer Function of the Tank Gain (Eq. 1)

From the equivalent circuit, the tank gain is the magnitude of its transfer function:

$$\boxed{K(Q,m,F_x) = \left|\frac{V_{o\_ac}(s)}{V_{in\_ac}(s)}\right| = \frac{F_x^2(m-1)}{\sqrt{(mF_x^2-1)^2 + F_x^2\left(F_x^2-1\right)^2(m-1)^2Q^2}}} \tag{5}$$

This expression is determined solely by three **dimensionless parameters** Q, m, F_x. Each is defined as follows:

$$Q = \frac{\sqrt{L_r/C_r}}{R_{ac}} \quad \text{quality factor (represents how heavy the load is)} \tag{6}$$

$$R_{ac} = \frac{8}{\pi^2} \cdot \frac{N_p^2}{N_s^2} \cdot R_o \quad \text{reflected load resistance} \tag{7}$$

$$F_x = \frac{f_s}{f_r} \quad \text{normalized switching frequency (the control knob)} \tag{8}$$

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}} \quad \text{resonant frequency} \tag{9}$$

$$m = \frac{L_r + L_m}{L_r} \quad \text{ratio of total primary inductance to resonant inductance} \tag{10}$$

Physical meaning of the three parameters: F_x: the ratio of the present switching frequency to the resonant frequency; the only quantity adjustable in real time. Q: varies in real time with the load. **The heavier the load → the smaller R_o → the smaller R_{ac} → the larger Q**. The coefficient $8/\pi^2$ comes from the FHA, and $N_p^2/N_s^2\,R_o$ is the reflection of the secondary-side load. m: the smaller m → the smaller L_m → the stronger the boost capability and the more flexible the regulation, but the larger the magnetizing circulating current and its loss; the larger m, the opposite.

4.2 The Gain Curves (Plots of Eq. 1)

Plotting K against F_x, with each curve corresponding to one Q (one load), gives Fig. 4.

(a) All load curves intersect at $F_x = 1$ (i.e. $f_s = f_r$) at a gain of 1. This is because at the series resonant point the impedances of L_r and C_r cancel, the series branch approaches a short circuit, and the input voltage is applied almost unchanged to the output node, so $K = 1$; moreover, **this conclusion is independent of the load (Q)**, so all curves are forced through $(1, 1)$. This point is called the **resonant point** and is the natural anchor of the design: the gain does not drift with load, the tank is purely resistive, the circulating current is minimal, and the efficiency is highest. In design one precisely **chooses the turns ratio in reverse so that the rated input/output condition falls at this point**. Low-Q curves correspond to light load, high-Q curves to heavy load.

(b) Every curve has a peak. The peak lies between two resonant frequencies: the higher f_r (L_r with C_r) corresponds to $F_x = 1$; the lower one,

$$f_{r2} = \frac{1}{2\pi\sqrt{(L_r + L_m)\,C_r}} \tag{11}$$

corresponds to the case where none of the secondary diodes conduct and L_m also participates in the resonance. The peak divides each curve into two operating regions (the orange operating regions):

Table 2: Inductive region vs. capacitive region

Region	Location	Current phase	Consequence
Inductive	**right** of the peak (higher frequency)	current **lags** voltage	ZVS achievable (safe)
Capacitive	**left** of the peak (lower frequency)	current **leads** voltage	ZVS impossible (dangerous)

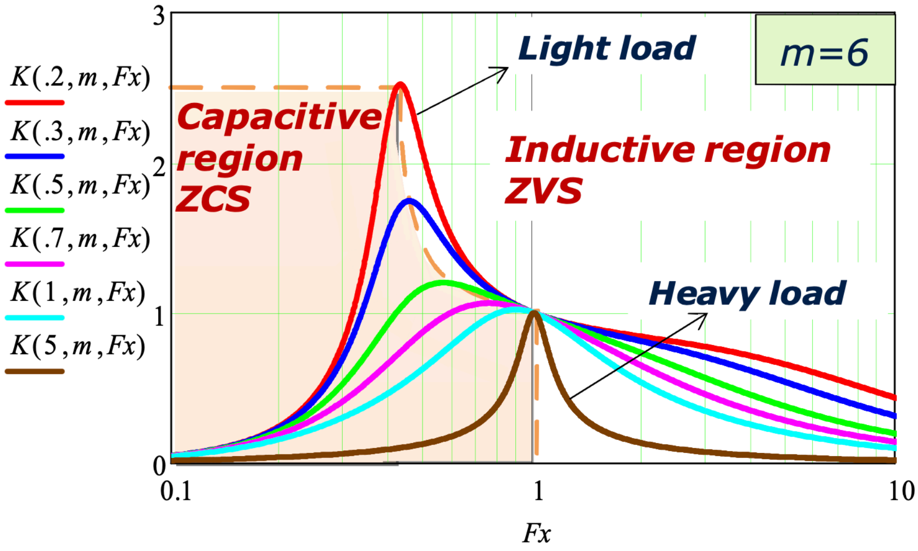

Figure 3: Tank gain K versus normalized frequency F_x ($m = 6$)

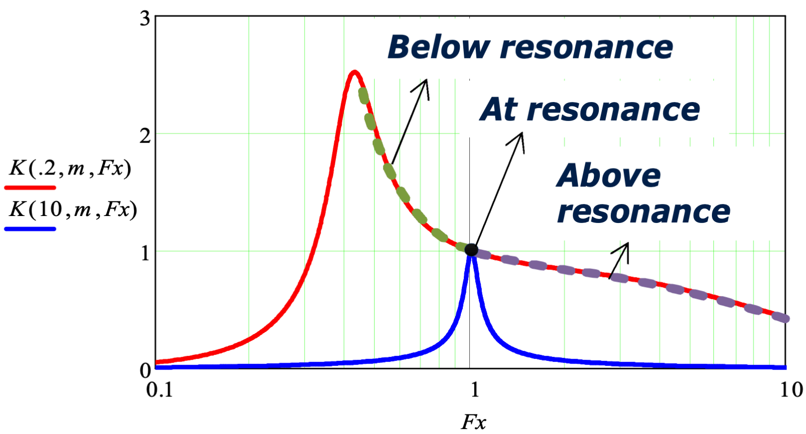

Figure 4: The operating regions formed by the resonant tank

4.3 Why Operation Must Stay in the Inductive Region: ZVS

One principle of LLC design is: **over the entire input-voltage × entire load range, the operating point must remain in the inductive region (right of the peak)**. The reason lies in soft switching:

Inductive region (current lags) — achieving ZVS:

1. The resonant current lags the bridge-leg voltage; at the instant the MOSFET turns off, the inductor current cannot change abruptly and keeps flowing in its original direction;
2. This current discharges the output capacitance C_{oss} of the device **about to turn on** (while charging the one just turned off);
3. When C_{oss} has discharged to $V_{ds} \approx 0$, its body diode freewheels first;
4. Only then is the turn-on signal issued → the device turns on at $V_{ds} = 0$, and the **turn-on loss ≈ 0 (ZVS)**.

Capacitive region (current leads) — dangerous:

1. The current leads, so before the MOSFET has turned off the current has already reversed and flows into the body diode;
2. When the complementary device turns on, the conducting body diode is **forced into reverse turn-off (hard commutation)**;
3. This produces a large reverse-recovery current spike, loss, noise and EMI, and in severe cases device failure.

The ZVS boundary = the peak of the gain curve. The essence of "find the minimum normalized frequency F_{x_min}" in the Section 3 design procedure is to pin the lower frequency limit at the **peak of the heaviest-load curve**, ensuring that no operating condition can slip into the capacitive region.

5. Analysis of the Operating Modes

Since the LLC is frequency modulated, depending on the input voltage and load, the converter has three modes: $f_s = f_r$, $f_s > f_r$, and $f_s < f_r$. Although there are three modes, within one switching period they are assembled from only **two basic actions**: **power delivery** and **freewheeling**.

5.1 Two Key Currents

Understanding the operating modes requires distinguishing two currents:

- **The resonant current I_{Lr}**: approximately **sinusoidal**, determined by the resonant tank.
- **The magnetizing current I_{Lm}**: approximately a **triangular wave**. The reason: during power delivery the secondary conducts, the output voltage is reflected to the primary and **clamps** across L_m (about $\pm nV_o$, a constant); a constant voltage across an inductor → a linearly changing current → a triangular wave.

The core relation: the current actually sent through the transformer to the secondary and to the load is $I_{Lr} - I_{Lm}$.

- $I_{Lr} \neq I_{Lm}$: current flows to the secondary → **power delivery** is in progress;
- $I_{Lr} = I_{Lm}$: the secondary current is zero and the diodes turn off naturally → **freewheeling** begins.

5.2 The Two Basic Actions

Power delivery (Fig. 5): The resonant tank is excited by $+V_{in}$ or $-V_{in}$, and I_{Lr} resonates sinusoidally; during power delivery the magnetizing-inductance voltage is the positive/negative reflected output voltage, so the magnetizing current charges/discharges accordingly; the difference between I_{Lr} and I_{Lm} is sent to the secondary through the transformer and rectifier, delivering energy to the load.

Freewheeling (Fig. 6): Occurs only after power delivery, and only **when I_{Lr} falls back to catch up with I_{Lm} — this happens only when $f_s < f_r$**. At this point the secondary rectifier current reaches zero and the rectifier disconnects; L_m is released from the clamp and can enter a second, lower resonance together with L_r and C_r; because $L_m \gg L_r$, this resonant frequency is far below f_r, so during freewheeling the primary current changes very little and can be treated as approximately constant. This stage **delivers no energy to the load**; the current merely idles on the primary, forming a circulating current.

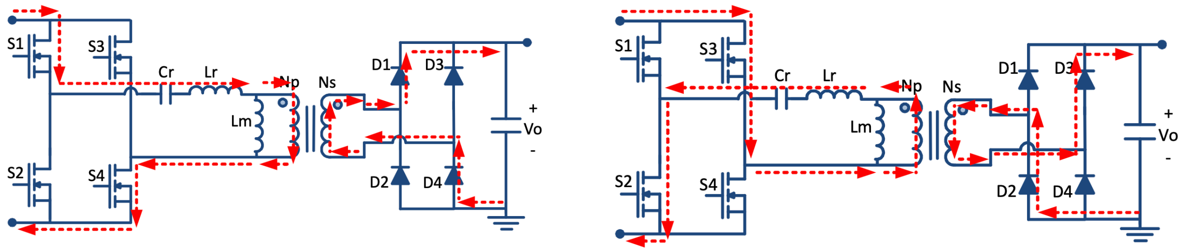

Figure 5: Equivalent circuit of the power-delivery action

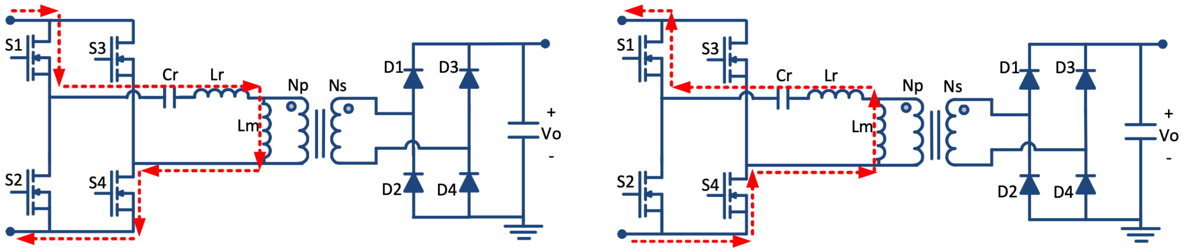

Figure 6: Equivalent circuit of the freewheeling action

5.3 Summary of the Three Modes

Mode 1: $f_s = f_r$ (the resonant point)

The two half-periods are equal in length, and the resonant half-period completes exactly within the switching half-period. At the instant the switching half-period ends, I_{Lr} has just fallen to catch up with I_{Lm} and the rectifier current has just reached zero — the secondary diodes achieve **zero-current turn-off (ZCS)** with no reverse recovery. At this point the tank gain is 1, there is no freewheeling and no interruption, and operation is optimal and most efficient. **In design, the transformer turns ratio is chosen so that the rated input/output condition falls at this point.**

Mode 2: $f_s > f_r$ (above resonance, step-down)

The switching half-period is **shorter than the resonant half-period**. The resonance has not yet completed (I_{Lr} has not returned to I_{Lm}) when the next switching half-period starts ahead of time and **interrupts** it, so each half-period contains only **partial power delivery**. Consequences: the primary MOSFET current is still large at turn-off → **increased turn-off loss**; and the secondary rectifier diode undergoes **hard commutation**. This mode is used when the **input voltage is high** and a step-down is needed.

Mode 3: $f_s < f_r$ (below resonance, step-up)

The switching half-period is **longer than the resonant half-period**. The resonance **completes early**, I_{Lr} catches up with I_{Lm} early, and **freewheeling** then begins and lasts until the switching half-period ends. During freewheeling there is **circulating energy** on the primary but no power transfer, leading to **increased conduction loss**. This mode is used when the **input voltage is low** and a step-up (gain > 1) is needed.

Secondary-side ZCS is a hidden advantage of the LLC: when $f_s \leq f_r$, the secondary diode currents all fall naturally to zero before turning off, with no reverse-recovery loss, keeping the output side efficient as well. Only $f_s > f_r$ breaks this advantage, so most designs try to avoid operating above resonance for extended periods.

I_{Lm} transfers no energy, yet it secures primary-side ZVS: the triangular magnetizing current is responsible, within the switching dead time, for charging/discharging C_{oss} and smoothing out V_{ds}. It **depends only on L_m and the voltage, not on the load**, so as long as L_m is small enough (the magnetizing current is large enough), **primary-side ZVS can be achieved even**

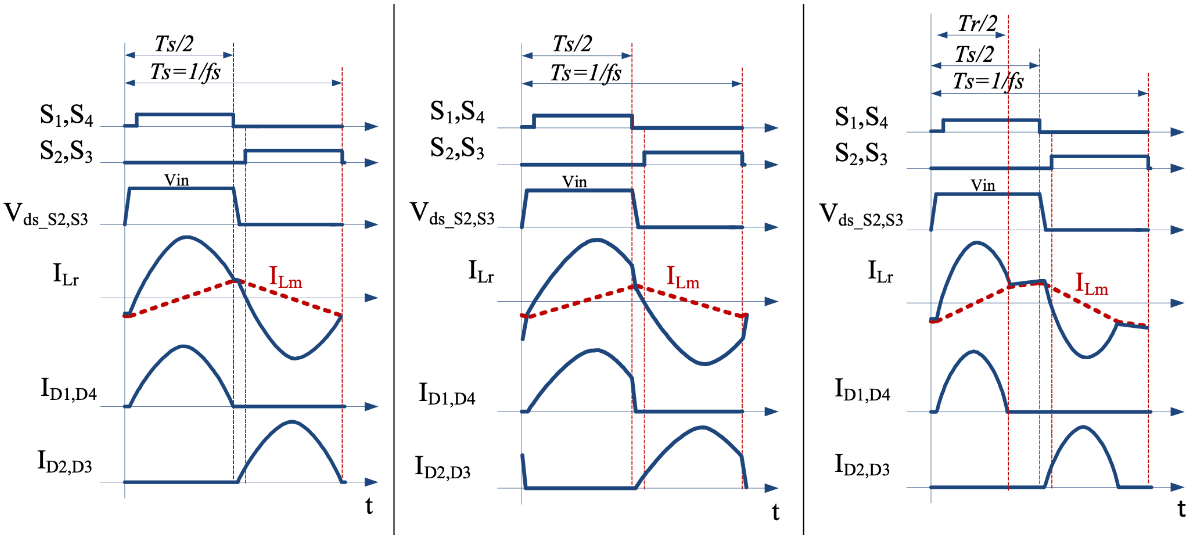

Figure 7: Comparison of the key waveforms in the three modes

at no load. This is exactly why the subsequent choice of m (i.e. the choice of L_m) must be made with care: too large an L_m will cause loss of ZVS at light load.